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The Algebras of Design

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Abstract. Shapes made up either of points, lines, planes, or solids belong to algebras that separately and in Cartesian products provide the main objects and devices used in shape grammars.

Designers work with descriptive devices of many kinds. These may be spatial or symbolic. Chief among them are drawings and models, and numbers and words. Drawings and models are central to practice. They are manipulated singly, togehter, and with numbers and words to express ideas about form, function, and the use of technology and material, and, in this way, to create designs.

The properties of drawings and models and their use in practice can be studied in terms of shapes and computations with them.

Shapes are formed in arrangements of points, lines, planes, and solids. No matter which one of these basic elements is involved, shapes have parts that may be distinguished in any way whatsoever, they may be combined by putting several together or by taking one away from another, and they may be changed into similar ones by moving them, reflecting them, or varying their size. These three properties determine algebras of shapes, each ordered by a subshape relation, and closed under operations of sum and difference, and transformations that are Euclidean. This is illustrated nicely when shapes are formed with straight lines.

Lines of this kind have distinct endpoints that fix their boundaries, and finite, nonzero length; they are related if one is embedded in another as a smaller or equal segment. Lines are arranged to form shapes. These arrangements are given by finite sets. Not any set of lines will do, however, if different sets of lines are meant to determine different shapes. Alternative sets of lines form the shape in Fig. 1, say, seven lines: the shared side and the six other sides of the squares, or five lines: the two horizontals and the three verticals. The lines arranged to form a shape must be chosen with care to be sure that a single shape is not mistaken for many. Maximal lines are used for this purpose. Maximal lines may be colinear if they are separated by a gap, but otherwise they may touch or intersect. Every shape is formed uniquely by a set of maximal lines, and no other set of lines that forms the shape contains fewer lines or longer ones.

As a set of maximal lines, a shape has a limited number of parts given by the members of the set or its subsets. But the maximal lines in the shape have other lines embedded in them that may be maximal in combination. In this case, sets of these lines form shapes that are parts of the shape, and that may be distinguished in it. The subshape relation fixes this idea: one shape is a part of another shape if evey maximal line in the first is embedded in some maximal line in the second. The shape in Fig. 1 is formed by a set of five maximal lines: two horizontals and three verticals. These contain the many parts of the shape, including a rectangle and two squares. The rectangle combines maximal lines in the set and is formed by a subset of these. Neither of the squares, however, is formed by any such subset; both have maximal lines that are not in the shape but that are embedded in maximal lines that are.

The subshape relation allows for shapes to be divided into parts in any way that may be convenient. In a reciprocal manner, shapes are combined to become parts in sums and to take away parts in differences. Intuitively, sum corresponds to drawing several shapes as one, and difference to erasing the parts of one shape that are shared by another. The definitions of both operations follow from the subshape relation, but are more readily given by reduction rules. These further provide alternative characterizations of maximality and the subshape relation. The lines in a set are maximal if no two can be combined using the reduction rules for sum. And

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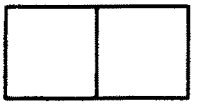


Fig. 1.

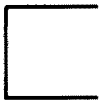


Fig. 2.

one shape is a part of a second if their sum and the second are the same.

The reduction rules for sum and difference are needed to maintain the correspondence between shapes and sets of maximal lines. This is easy to see when these shape operations are compared with the corresponding set operations of union and relative complement.

The two squares in the shape in Fig. 1 are formed by a set of four maximal lines apiece, and, in union, these sets produce a set of seven lines: four horizontal sides and three vertical ones. However, the horizontal sides in this set are not maximal lines in the shape, but make them when combined in certain pairs. The reduction rules for sum combine lines to produce maximal lines; they are applied to the set of seven lines to obtain the set of five maximal lines that forms the shape.

Difficulties of another kind arise for difference. The difference of the shape in Fig. 1 and its rightmost square is shown in Fig. 2. However, the set of three maximal lines that forms this shpae is different than the set of three lines produced in the relative complement of the set of maximal lines for the shape in Fig. 1 and the set of maximal lines for the square. The horizontals in the relative complement have the horizontal sides of the square embedded in them, but no line embedded in any of these sides is embedded in either of the horizontals in the difference. The reduction rules for difference are applied to the set of maximal lines for the shape in Fig. 1 in terms of the set of maximal lines for the square to obtain the set of maximal lines that forms the shape in Fig. 2.

Shapes formed by sets of maximal lines, the subshape relation, and the operations of sum and difference determine an algebra that satisfies the axioms for a Boolean ring. The algebra has a zero given by the empty set, and is without the benefit of a unit that would violate the requirement that shapes be finite sets of lines of finite length. When the transformations are used, a more elaborate algebra, U_1, is defined with resources sufficient to frame rules for computations with shapes, and to be useful in practice. And in fact, with the trivial exception of the set containing the zero only, U_1 has no proper subset with this algebraic structure.

The definition of U_1 is common to other algebras of shapes formed by arrangements of points, planes, and solids. (Planes and solids are connected and may have holes; if wanted, they can be produced from triangles and tetrahedra joined by edges and faces.) These algebras are organized in a hierarchy in Table 1.

Each algebra satisfies the axioms for a Boolean ring: the empty set is the zero and there is no unit except in U_0 when it is zero-dimensional, and the algebra is closed under the transformations. The algebra is named and placed in the hierarchy according to the dimension of its basic elements, either points, lines, planes, or solids. This dimension is fixed. But the dimension of the algebra itself may be given in any way that use demands. This dimension varies with the transformations, being free to equal or to exceed the dimension of the elements of the algebra. For example, three-dimensional algebras with shapes formed by sets of points, lines, planes, or solids may be used as desired. And algebras of more than three dimensions are sometimes useful, especially where computations with multiple shapes are involved.

The main details of the algebras in Table 1 depend on the subshape relation and the operations of sum and difference. In U_0, points are embedded

Table 1.

	Objects	Basic elements		Boundaries
Algebra	Shapes determined by finite sets of maximal	Defined in dimension	Have finite, nonzero	Every basic element has a boundary that belongs to
U_0	Points	0	—	—
U_1	Lines	1	Length	U_0
U_2	Planes	2	Area	U_1
U_3	Solids	3	Volume	U_2

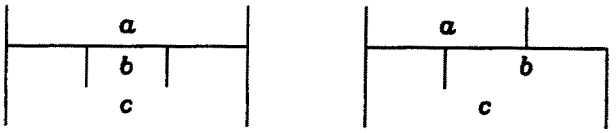


Fig. 3.

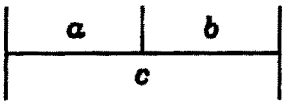


Fig. 4.

one in another if and only if they are the same. This allows subshape to coincide exactly with the subset relation, and allows sum and difference to coincide with the set operations of union and relative complement. In the other algebras, however, embedding plays a central role in relating basic elements, and defining the subshape relation and reduction rules for sum and difference. And, as they do for U_1, reduction rules also provide alternative characterizations of maximality and subshape.

Reduction rules can be framed in a general sort of way for both sum and difference. The reduction rules for sum apply recursively to sets of basic elements; they combine two elements a and b of U_{i+1} to make an element c whenever (1) a and b can be embedded in an element, and either (2) an element is embedded in both, or (3) this is not so and the boundary of a and the boundary of b share a part formed by one or more elements of U_i. Condition 1 requires colinearity, coplanarity, or solids that are defined in a common three-dimensional space, and conditions 2 and 3 require that lines, planes, or solids are joined either because they overlap, as shown for lines in Fig. 3, or because their boundaries do, as shown in Fig. 4. The element c is then defined by the boundaries of a or b or both, and has a and b embedded in it.

In an anlogous fashion, the reduction rules for difference apply recursively to sets of basic elements in terms of some given set of basic elements, either to remove an element, or to replace it by one or more other elements. On the one hand, an element a in a set to which the rules apply is embedded in an element b in the given set, and on the other hand, this is not so, and an element embedded in b is embedded in a. This is clear for lines, where a is removed in Fig. 5, and replaced by elements c, or c and d in Fig. 6. Every new element is defined by the boundaries of both a and b, so that it is embedded in a, but has no element embedded in b embedded in it.

The algorithmic details of reduction rules depend on how basic elements and the embedding relation are represented. In U_1, lines have a simple representation, each as a string encoding the coordinates of the points that bound it and equations that describe it. A lexicographic ordering of strings then determines the embedding relation, and can be exploited easily to check the subshape relation and to carry out sums and differences. This kind of approach in which an ordering is imposed on strings that represent basic elements in terms of boundaries and descriptions also appears to be appropriate for planes and solids in U_2 and U_3. However, additional techniques—many pioneered in solid modeling—may be needed to complete the job efficiently.

In practice, drawings and models may not correspond to simple shapes—ones formed out of basic elements of a single kind— but suggest instead that shapes involve a mixture of points, lines, planes, and solids. And several drawings, and drawings and models, may be used together in coordinated ways, perhaps as separate descriptions of the different components in an assembly, or as multiple descriptions of the same thing, say, building plans, sections, elevations, and models. These important extensions lead to other algebras. They are readily defined when the algebras in Table 1 are combined in Cartesian products, and share their properties as Boolean rings closed under the transformations. Objects in the algebras are compound, connecting shapes of various kinds. The shapes in each object may be taken all at once as in a single drawing or model, or in relationship to one another as is done when multiple drawings and models are used together. Complex shapes are thereby formed on the one hand, and shapes are coordinated for a common purpose on the other.

Numbers and words are also used with drawings and models. Such symbolic devices among others may be combined with shapes to study this relationship. Two main approaches are open.

In the first approach, the algebras in Table 1 are augmented by associating labels or properties with basic elements. This is done in U_0 when labels from a specified set, say, the alphabet $a, b, c, \ldots$, are attached to points. Labels may be given simply to

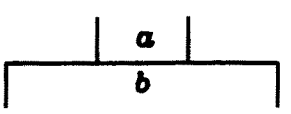


Fig. 5.

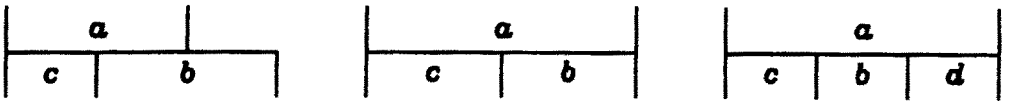


Fig. 6.

identify and classify points, or they may have a semantics allowing them to carry important information about shapes and other things. Either way, labeled points are ordered pairs to be arranged in sets to form an algebra in the manner of U_0. If the transformations keep labels the same in accordance with everyday custom, then the algebra and U_0 have common properties. The algebra may have other properties, too, in the event that labels attached to the same point are meant to combine.

In the second approach, alternative algebras are introduced to combine with algebras of shapes in Cartesian products, allowing for shapes and symbolic devices to be connected. For example, shapes—simple or complex ones with or without labels and properties—are described in decompositions. These are defined in a variety of ways, but most simply as sets of shapes that show how their sums are divided into parts of certain kinds. As sets of shapes, decompositions belong to a Boolean ring closed under the transformations. Every set in this algebra except the zero is a decomposition of some shape, and every shape has at least one decomposition and possibly infinitely many of them in the algebra. The algebra is valuable by itself, but is sometimes used with other devices. Shapes may be named to place them in categories, just as points are labeled in U_0. Or decompositions may be organized more finely, so that the shapes in each are ordered hierarchically, or in some other way.

All of the algebras mentioned so far are defined for objects that are shapes, or that are made from shapes in simple set-theoretic constructions. And all of these algebras structure sets of objects in exactly the same way: there is an ordering relation, $\leq$, on objects and their parts that is framed in terms of subshape and subset, an operation, $+$, to put objects together using sum and union, an operation, $-$, to take one object away from another using difference and relative complement, and transformations t that are Euclidean. This uniformity provides for a general approach to computation that is characterized completely in the shape grammar formalism.

A shape grammar is used in an algebra to perform computations by following rules. In this way, objects in the algebra are produced, and a language—a subset of the algebra—is defined and explored. Every shape grammar works in a uniform manner, whether in the algebras in Table 1, in versions of these algebras augmented with labels and properties, in algebras containing decompositions, or in algebras formed in Cartesian products. For example, early formal work on shape grammars was carried out in repeated Cartesian products of U_1. And most of the shape grammars that have been developed in architecture to characterize building styles and types use rules for computations in the Cartesian product of U_1 and a version of U_0 in which labels are attached to points.

In a shape grammar, any pair of objects A and B defines a rule $A \rightarrow B$. The rule applies to an object C in a two-stage process involving a transformation t. The transformation is used in both stages, once with the relation $\leq$ to distinguish some part of C, and then again with the oeprations $+$ and $-$ to replace the part that has been picked out. In exact detail, the rule $A \rightarrow B$ applies to C if the formula

$$\text{there is a } t \text{ such that } t(A) \leq C \qquad (1)$$

is satisfied. In this case, a new object may be produced according to the formula

$$(C - t(A)) + t(B) \qquad (2)$$

The rules in a shape grammar apply recursively in accordance with Eqs. (1) and (2) to determine the series of objects in a computation. The mechanism described by the formulas exploits the full resources of the algebra in which the grammar is defined, and offers many exciting possibilities for computations. Some of these arise effortlessly when the objects in rules are connected directly by the relation $\leq$.

For example, it is evident that an application of a rule $A \rightarrow B$ can be followed by another if there is a transformation that makes A a part of B. Fractals like the one in Fig. 7 are produced using rules of this kind in the algebra U_1, and so are symmetrical patterns like the one in Fig. 8. In both of these cases, rules take the form $A \rightarrow \Sigma t(A)$, allowing for A to be

Fig. 7.

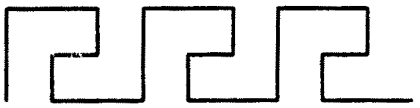


Fig. 8.

replaced with a sum of several transformations of itself, either with or without changes in scale.

But there is a lot more to shape grammars than fractals and symmetrical patterns, and all of it is of greater interest and use. The idea of connecting the objects in rules by the relation $\leq$ can be extended profitably, and novel ways of defining and using rules can be introduced that depend on other aspects of Eqs. (1) and (2). In all of this, emergence, indeterminacy, and symmetry play important roles.

Everyday accounts of what to expect in computations usually begin by assuming two things. First, every object in a computation has a unique description in terms of standard parts that are fixed before the computation begins and kept constant as it proceeds. And second, these parts are independent in combination, being the only ones that are recognized and used as rules are applied. Neither of these assumptions, however, must pertain to computations in shape grammars. An object in a computation may not have a unique description in terms of standard parts, or any description of this sort at all. And no one description of an object need restrict the way rules apply.

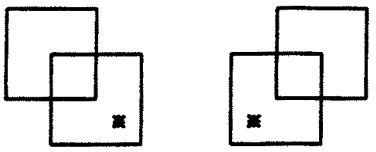
In a computation in a shape grammar, the parts of an object and their consequent possibilities for organization are indefinite, being contingent on rules and transformations. The parts of the object may be established in a different way every time a rule is applied, and need not be anticipated. They may vary discontinuously—in the manner of gestalt shifts, figure-ground reversals, and ambiguous figures—with respect to a single rule and with respect to several different ones. This instability is crucial if computations with shapes and objects made from them are to correspond to the way drawings and models are used in practice. It allows for emergent properties to be recognized, and to be exploited freely.

Emergence is illustrated nicely in a simple shape grammar in the algebra U_1. The two rules in Fig. 9 both translate a chevron along its axis of symmetry, but in different directions. (The point marked by * in these rules is given for reference only to indicate the relationship between the chevron in the left side of a rule and the chevron in the right side. The device may be used generally to relate the components of objects, or objects in rules or in computations.) The rules apply to connect the shapes in Fig. 10 in a computation. This is surely impossible, and not because these shapes appear to be pairs of intersecting squares. With respect to the rules, each of the shapes consists of two chevrons back to back with a common axis. However, for the shapes together, these axes are orthogonal, with the second shape a 90° rotation of the first. How can rules that merely translate a chevron along a fixed axis carry out this complete change in orientation? Once the axis of a chevron is given, there is no way to perform the required rotation.

The answer is easy to see in the two-step computation in Fig. 11 in which the rules in Fig. 9 are used one each. In step 1, the first rule is applied to the first shape in Fig. 10 to translate the left-most chevron. A new shape is produced that can be described in two ways with respect to the second rule and possible ways of applying it. These descriptions compete with one another, and are incompatible for different applications of the rule. One description is expected. In it, two chevrons are distinguished, as shown in Fig. 12, that correspond to the two chevrons in the original shape. Neither of these chevrons, however, is recognized in the other description. Two emergent chevrons are distinguished instead, as shown in Fig. 13. In step 2, the second rule is applied to translate the left-most of these chevrons to produce the second shape in Fig. 10.

Examples like this one—and there are many of them—are engaging in their own right, but provide more than mere entertainment. They manifest the flexibility of Eqs. (1) and (2) whenever they are used in algebras in which the embedding relation is involved fundamentally. An object in a computation is changed into another one by applying a rule with respect to some description. This description, however, does not impose restrictions on other rule applications to the object itself or to successive objects in the computation. Rules may be defined and used with complete freedom, because the necessity

Fig. 9.



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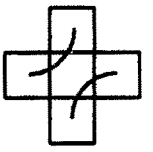

Fig. 12.

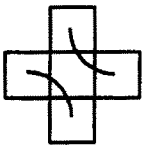

Fig. 13.

to know how objects have been described or have come to be never intrudes. As a result, the history of objects may play a role by choice, or may be ignored completely. A designer is welcome to enter a computation at any time with different ideas given in rules, and several designers working independently or in concert with different rules expressing their separate points of view can contribute to the same computation.

Of course, whenever conventional descriptions are established for objects, they may be used in computations in shape grammars. This is accomplished in a number of ways. Very simply, if two algebras are combined in a Cartesian product, then a reciprocal relationship can be formed in which rules defined in one algebra control the use of rules defined in the other. The rule in Fig. 14 applies in the algebra U_0 to replace any point with any different point. The rule can be combined with the rules in Fig. 9 in the Cartesian product of U_0 and U_1, as shown in Fig. 15. In these new rules, a chevron and a point are associated, so that a single chevron and translations of it are distinguished in computations beginning with the chevrons and the point in Fig. 16. Thus, emergent chevrons are no longer a possibility.

This result, however, is not guaranteed for every combination of the rule in Fig. 14 with the rules in Fig. 9, or in every computation in which the rules in Fig. 15 are used. For example, emergent chevrons are still a possibility in computations beginning with the chevrons and the point in Fig. 17 if the rules in Fig. 18 are defined and used. And this is also true in computations beginning with the chevrons and the points in Fig. 19 when the rules in Fig. 15 are used again. In a shape grammar, the parts of objects are established within computations and cannot be fixed from without.

In everyday accounts of computations, other things are assumed, too. It is taken for granted that rules apply in finitely many ways only, no matter what objects are involved. This may be the case in shape grammars, but it is not necessarily so. By itself, the rule in Fig. 14 is indeterminate: it applies to every shape in U_0 that has a point in it under infinitely many transformations involving rotations and changes in scale. And the rule in Fig. 20 is also indeterminate in use, as illustrated in Fig. 21. The rule applies to every shape in U_1 that has a line in it under infinitely many transformations involving translation and scale. These examples make the pattern clear: if a rule $A \rightarrow B$ is defined in an algebra in Table 1, then it is indeterminate in application to every shape with at least one maximal element, whenever the maximal elements in A can be embedded in a single basic element of the algebra.

The detailed conditions for indeterminacy, however, are more complicated, and vary from algebra to algebra and from dimension to dimension. The algebras U_0 and U_1 provide good illustrations in three dimensions. For a rule $A \rightarrow B$, indeterminacy depends entirely on the properties of A alone. The rule is indeterminate in application, at least in some cases, if and only if the following statements are satisfied. In the algebra U_0, this condition fails:

1. There are three points in A that are not colinear.

In the algebra U_1, one of these conditions fails:

2. There are two maximal lines in A that are skew, and

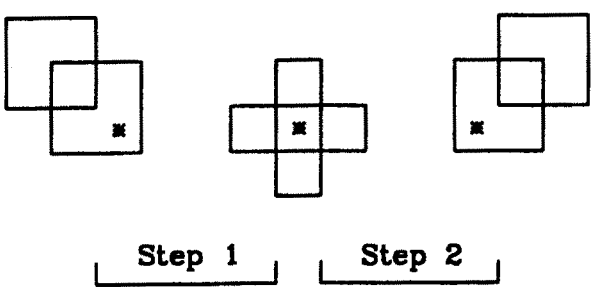

Fig. 11.

Fig. 14.

Fig. 15.

3. There are three maximal lines in A that are not colinear with a common point, and it is not the case that two of these lines are colinear or that all of them are parallel.

And in the Cartesian product of U_0 and U_1, either this condition fails:

4. A point and a maximal line in A are not colinear,

or one of the previous conditions, 1, 2, or 3, does.

It is easy to see that each of the conditions 1, 2, and 3 implies the other condition 4. Also, because all of the conditions invovle several maximal elements, they suggest that rules that are indeterminate in either U_0 or U_1 need not be indeterminate in combination in a Cartesian product of either one or both of these algebras. Thus, for example, the rules in Figs. 14 and 20 combine in the Cartesian product of U_0 and U_1 to form the rule in Fig. 22 that is not indeterminate in use. The rule is such that a transformation makes its left side a part of its right side, allowing for objects like the one in Fig. 23 to be produced in computations beginning with the point and the line in Fig. 24. This way of limiting indeterminacy in Cartesian products works for all of the algebras in Table 1, and is especially useful as indeterminacy becomes harder and harder to avoid in U_2 and U_3.

Conditions 1–4 have other important uses, too. In particular, they provide the basis for an algorithm to enumerate all of the transformations under which a rule $A \rightarrow B$ that is not indeterminate in use applies to a shape C in U_0 or U_1, or to an object C in their Cartesian product. And as a special case, if A and C are the same, then the algorithm determines the symmetry group of A. The idea behind the algorithm is straightforward. The maximal elements distinguished in conditions 1–4 serve as registration marks to define possible transformations, where marks in A and any corresponding marks in C register via identity if they are points and embedding if they are lines. The subshape relation is then used to decide whether these transformations produce the desired results.

There may be a tendency to find indeterminacy bothersome. This, however, would be a mistake. Rules that are indeterminate in application provide important advantages in shape grammars. They introduced possibilities for variation that are not ordinarily expected from rules, when each has a left and a right side in a fixed relationship and applies in accordance with Eqs. (1) and (2). It comes as no surprise that rules are applied in the algebra U_1 to produce shapes like the one in Fig. 25, because squares are inscribed in squares in a common relationship. But rules to produce shapes like the one in Fig. 26 in which this idea is generalized, so that quadrilaterals are inscribed in quadrilaterals, are not so easy to find. In this case, the relationship between two touching quadrilaterals can vary arbitrarily, and no quadrilateral need be a transformation of any other. Nonetheless, shapes like the one in Fig. 26 are produced by rules—very simply in the Cartesian product of U_1 and a version of U_0 augmented with labels—that are some indeterminate in application.

There are other ways to introduce possibilities for variation in shape grammars. The preferred approach begins with schemata to define shapes in the algebras in Table 1. A shape schema $A(x)$ is a set of varaibles x. Basic elements of an algebra are assigned to these variables by a function g, and may be required to satisfy certain conditions. Reduction rules for sum are then applied to the set $g(A(x))$ defined in this way to obtain the maximal elements that form a shape in the algebra. In the algebra, any

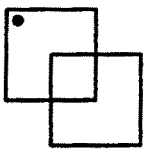

Fig. 16.

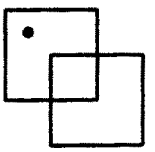

Fig. 17.

Fig. 18.

pair of shape schemata $A(x)$ and $B(x)$, with or without shared variables, may be joined in a rule schema $A(x) \rightarrow B(x)$. A function g assigns basic elements to all of the variables in this schema, so that any required conditions are satisfied. And once reduction rules ensure that elements are maximal, a rule is defined that may be applied in the usual way in accordance with Eqs. (1) and (2). In effect, this allows for shapes and their relationships to vary within rules, and extends the transformations under which rules apply. It further provides a straightforward means to express intuitive ideas about shapes and their use in computations. For example, injunctions such as "inscribe quadrilaterals in quadrilaterals" are now directly available in rules obtained from a single schema in the algebra U_1. And the same techniques work in algebras involving those in Table 1, and labels and properties, decompositions, and Cartesian products, when variables take appropriate kinds of values.

Another aspect of rules in shape grammars deserves attention. A rule $A \rightarrow B$ that applies to an object C to replace some part of it may do so in several distinct ways, depending on the symmetry properties of A and on how they are modified by B. This is easy to see in the algebra U_1. The rule in Fig. 27 applies to every equilateral triangle under six transformations corresponding to the separate rotations and reflections in the symmetry group of the equilateral triangle in the left side of the rule. But not all of these applications of the rule produce different results, as shown in Fig. 28. A rotation and a reflection in three pairs change the rhombus in the right side of the rule identically, thereby limiting what the rule can do. (Rotations are clockwise, and reflections are named in Fig. 28.)

The behavior of rules under different transformations can be described in a general fashion. Let transformations be equivalent with respect to an object whenever they change the object identically. If a rule $A \rightarrow B$ is such that this relation partitions the symmetry group of A into q classes in terms of B, then the rule can be used in q distinct ways. Further, if the symmetry group of A contains n transformations, then q divides n without remainder, in compliance with a famous theorem of Lagrange. The symmetry group of A has a subgroup containing n/q transformtions with left cosets given by the classes determined by B. (For the group operation "$\cdot$" and the transformations t and t', it is agreed that $t \cdot t'(A) = t(t'(A))$.) And conversely, there is a rule $A \rightarrow B$ for every subgroup of the symmetry group of A that behaves as the subgroup describes. The distinct uses of a rule $A \rightarrow B$ illustrate certain symmetry properties of A, and the symmetry properties of A serve independently to classify every rule in which A is a left side. This is shown nicely in Fig. 29 by giving a rule in the algebra U_1 for each

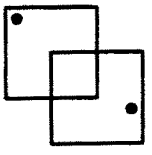

Fig. 19.

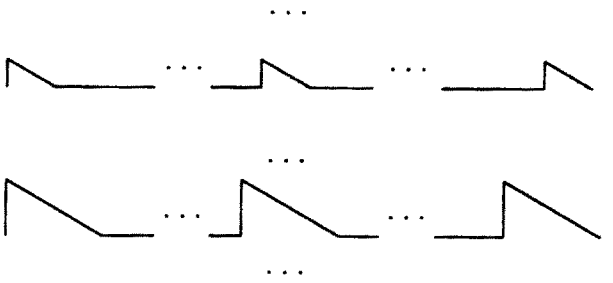

Fig. 21.

Fig. 20.

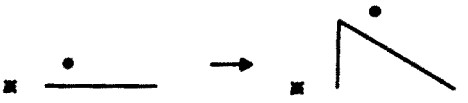

Fig. 22.

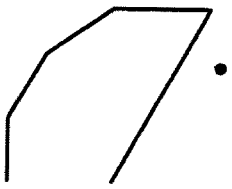

Fig. 23.

Fig. 24.

subgroup (with the exception of isomorphs) of the symmetry group of an equilateral triangle.

Sometimes, it is important to control the uses of a rule. And once again, Cartesian products provide a simple means of doing so. Any rule $A \rightarrow B$ in an algebra can have its left side augmented in a Cartesian product involving the algebra and another one. A new rule $A' \rightarrow B$ is defined with an asymmetric object A' that has A as a component. The transformations t that serve to apply the rule $A \rightarrow B$ to replace the same part of an object C in different ways also determine unique applications of the rule $A' \rightarrow B$ to replace different parts of the sum $C + \Sigma t(A')$. Every application of the rule $A \rightarrow B$ is sanctioned in the original algebra, but only selected applications of the rule $A' \rightarrow B$ need be sanctioned in the Cartesian product, depending on the transformations of A' that are actually summed with C. It is interesting to notice also that the rule $A \rightarrow \Sigma t(A')$

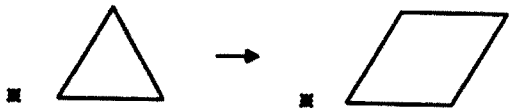

Fig. 27.

behaves in the manner described by a subgroup of the symmetry group of A, whenever the subgroup provides the transformations t used to make the right side of the rule.

The role played by Cartesian products to control the uses of a rule is clear in an example. The rule in Fig. 30 is defined in the Cartesian product of U_0 and U_1. The rule contains a point in its left side to break the symmetry of the equilateral triangle in the left side of the rule in Fig. 27. As a result, the rule in Fig. 30 applies selectively to the objects in Fig. 31, either once, twice, or three times.

But things are not always so simple in computations. A rule $A \rightarrow B$ may be applied in a computation only if there is a transformation that makes A a part of an object C formed in the sum of a part of the object that begins the computation and parts of transformations of the right sides of rules. If the uses of the rule are to be controlled by augmenting A in a Cartesian product, then the devices used for this purpose must be incorporated in C. This requires that the object that begins the computation and the right sides of rules also be augmented. The details of this process, however, depend on the properties of the rules and the computations involved, and vary widely from one case to the next.

Shape grammars are interesting in their own right

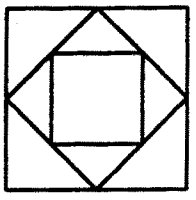

Fig. 25.

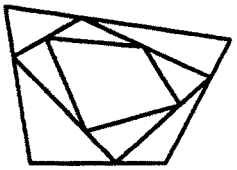

Fig. 26.

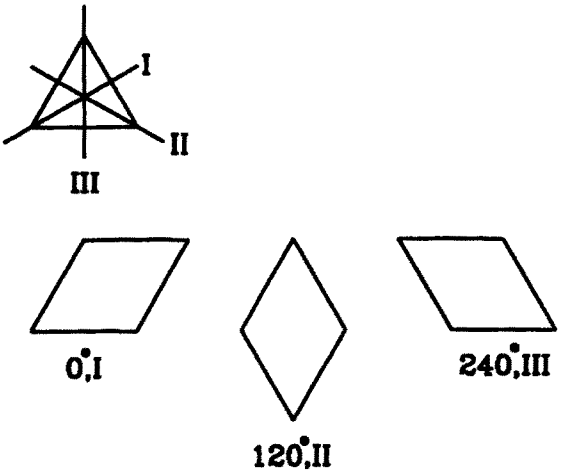

Fig. 28.

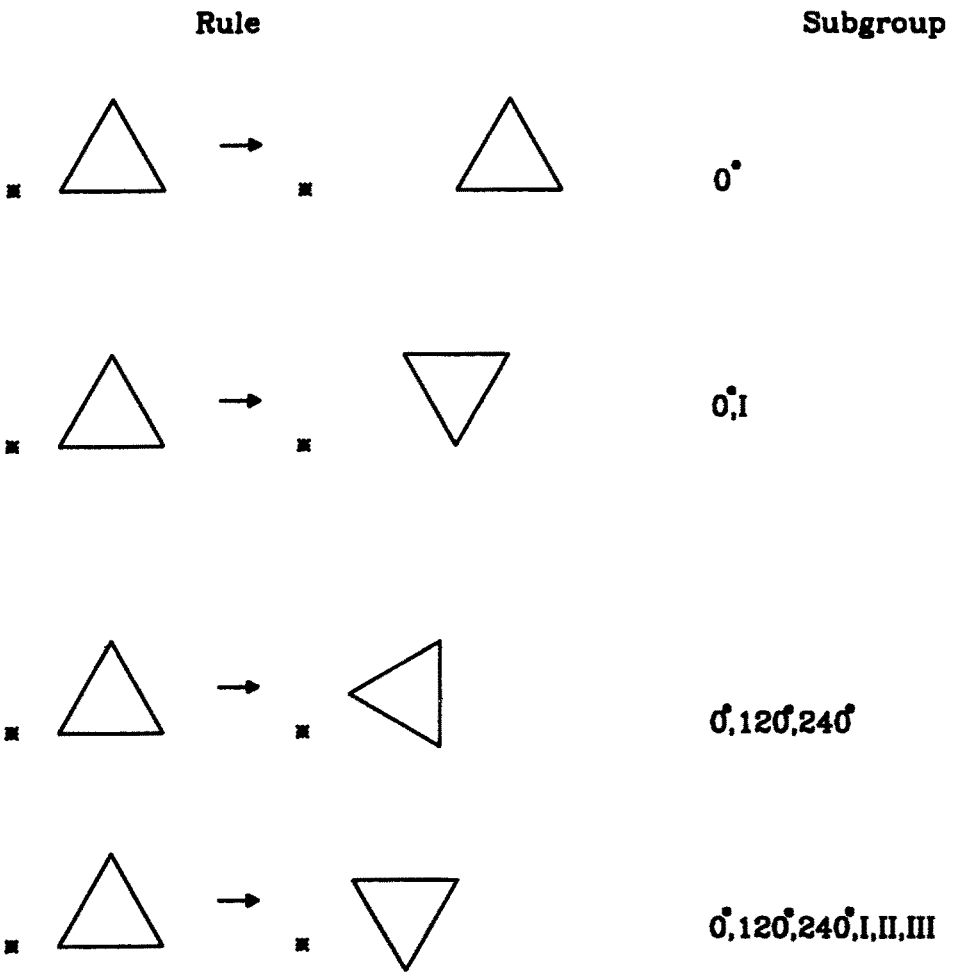

Fig. 29.

as formal systems with special properties. But their true worth comes in practice, where they define languages of designs. A language is a subset of an algebra containing objects that are produced in computations by applying the rules in a shape grammar. Languages are many and varied; they may be as inclusive as the algebra itself, or they may contain objects of certain kinds. It is in these cases that shape grammars and the languages they define are of the greatest importance.

In architecture, for example, shape grammars have been used to define languages for Palladio, Wren, Wright, and Terragni, and for vernacular styles from different periods and places. Shape grammars provide designs for churches, villas, houses, multi-storied structures, and buildings of other types, for ornament to decorate them, for furniture to fill them, and for gardens in which to put them. And new shape grammars are always forthcoming; they have been worked out from scratch to explore fresh ideas, and by combining or transforming shape grammars already in use to profit from past experience with established styles and types.

The definition of shape grammars in the algebras in Table 1 makes computations with shapes that have the properties of drawings and models a reality. And the definition of shape grammars in other algebras formed from the algebras in Table 1 leads

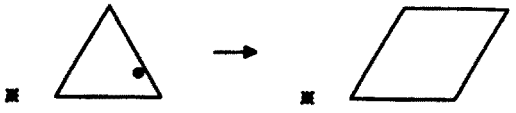

Fig. 30.

Fig. 31.

to computations with objects that combine shapes and symbolic devices. This gives shape grammars a central role in a more elaborate theory in which languages are seen as relations containing designs.

Shape grammars perform parallel computations in multiple algebras. In this way, shapes and symbolic devices available to specialists as descriptions in their separate domains of interest are connected in relationships to produce designs. Specialist domains have their own algebras and languages. And different domains interact and influence one another mutually when their algebras combine, and their languages are linked by shape grammars. This facilitates communication across domains as rules are defined and applied, and encourages a confluence of diverse intentions and ideas about form, function, and the use of technology and material. Shape grammars promise that designs can be produced in computations as successfully as they are in intelligent and creative practice.

The main details of this approach have been developed over the past 20 years. An important early account is given in Stiny (1975). Later discussions of special interest here include Stiny (1980; 1988; 1989; 1990*a*; 1990*b*). A representation of lines and its use in an algorithm to determine transformations to apply rules in the Cartesian product of the algebras U_0 and U_1 are considered in detail in Krishnamurti and Earl (1990). An important study of how properties, especially color, may be associated with planes and solids in shape grammars is presented in Knight (1989).

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